# Tools — Sub-Agents

Spawn focused sub-agents that run synchronously or in the background. Prefer sub-agents over doing work inline: delegate exploration, builds, and planning.

| Tool | Description |
|---|---|
| `new_agent` | Spawn a sub-agent (blocking or background). |
| `cancel_agent` | Cancel a running background sub-agent. |
| `list_agents` | List sub-agent tasks for the session. |
| `wait_agents` | Block until background tasks complete. |
| `agent_complete` | Terminal signal: the autonomous task is done. |

**System instruction:** "Prefer sub-agents over doing work yourself. Use `background: true` when spawning more than one."

Available agents: `explore` (fast, read-only codebase understanding), `build` (compile/test/fix), `plan` (implementation plans, read-only), `general` (full-capability fallback).

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## new_agent

Spawn a sub-agent to perform a focused task. **Both `agent` and `task` are required.** The agent is stateless — batch all related questions into one call with a comprehensive prompt.

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `agent` | string | yes | Agent name (`explore`, `build`, `plan`, `general`, or custom) | `"explore"` |
| `task` | string | yes | Specific prompt/instructions, including all needed context | `"Find all callers of X in src/"` |
| `background` | boolean | no | Run concurrently without blocking (default false). Use `true` whenever spawning more than one in the same response | `true` |

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `detached` | boolean | no | Only with `background: true`. Fire-and-forget: excluded from `list_agents`, not awaitable via `wait_agents`, completion NOT injected back into the session | `true` |
| `model` | string | no | Provider/model override | `"anthropic/claude-sonnet-4-20250514"` |

Concurrency: at most 32 background tasks per session; use `wait_agents` to free slots before spawning more.

A **detached** task (`background: true, detached: true`) runs and shows in the Agents panel but is invisible to the delegation surface: `list_agents` omits it, `wait_agents` (with or without `task_ids`) never returns it, and its completion is reaped without a chat injection while still appearing in `tasks_snapshot`. Because nothing reads its reply body, a detached prompt whose deliverable is a report MUST name a file to write; the run's full output is additionally persisted to `log/subagents/<task-id>.md` on completion. The TUI `/spawn` command is the user-facing equivalent.

**Example:**

```
new_agent agent="explore" task="Summarise architecture in crates/ragent-agent/src/agent/" backgro
wait_agents
```

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## cancel_agent

Cancel a running background task spawned with `new_agent background=true`. Not for team tasks.

**Arguments**

| Argument | Type | Required | Description |
|---|---|---|---|
| `task_id` | string | yes | Task identifier returned by `new_agent` |

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## list_agents

List sub-agent tasks for the current session with status and result summary; completed tasks include an `output_file` path with the FULL untruncated report under `log/subagents/<task-id>.md`.

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `status` | enum | no | Filter by `running` / `completed` / `failed` / `cancelled` | — |
| `task_id` | string | no | Get details for one task | — |

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## wait_agents

Block until one or more background tasks complete (preferred over polling `list_agents`). For team members use `team_wait` instead.

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `task_ids` | array | no | Specific tasks to await (default: all running) | `["task-abc"]` |
| `timeout_secs` | integer | no | Max seconds to wait before returning partial results | `300` |

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## agent_complete

**Terminal signal** — the current autonomous task is fully done. Ends the session loop and returns control to the user. Only call when all requested outputs have been produced.

**Arguments**

| Argument | Type | Required | Description |
|---|---|---|---|
| `summary` | string | yes | Concise summary of what was accomplished |

Do NOT pass `task_id`, `team_name`, `result`, or `output`; to mark a *team* task done, use `team_task_complete` inside a team.